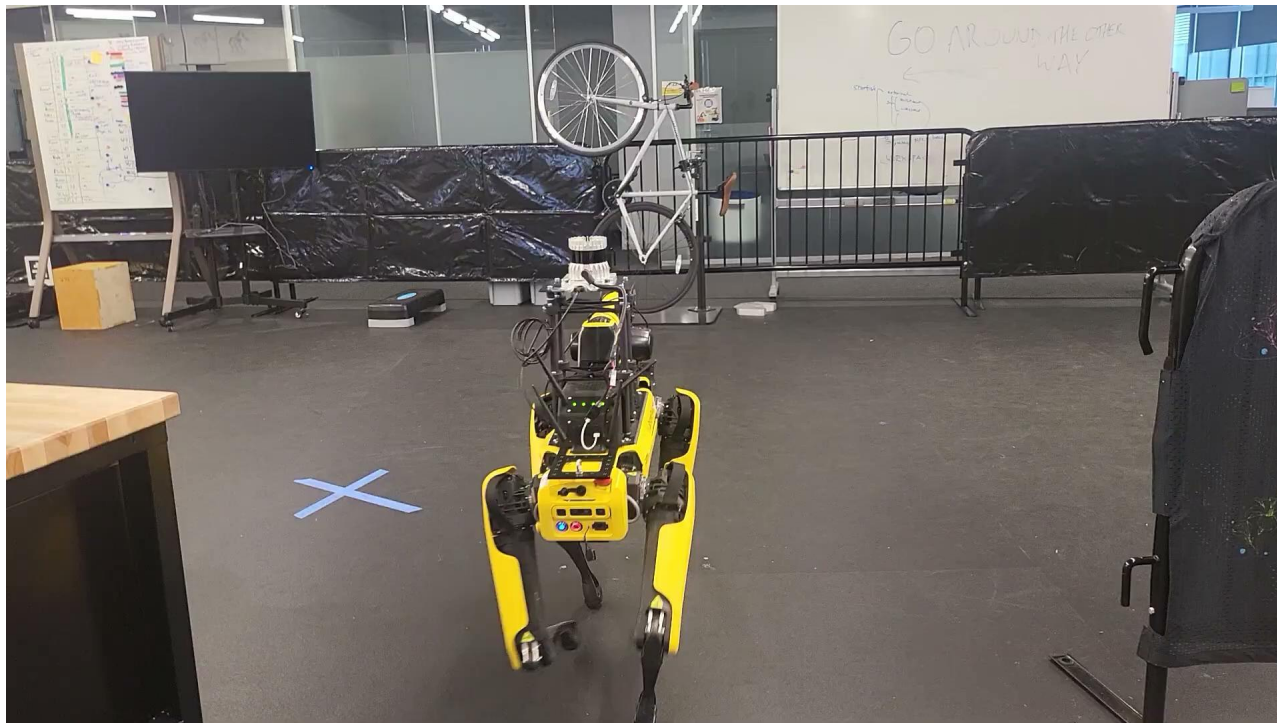


When Language Meets the Physical World: Semantic Representation for Robots

Presenter: Sonia Raychaudhuri
March 30, 2026



Move forward to the bicycle. Turn right, then move to the chair. Turn left, and stop near the potted plant.



Language must be grounded in perception

Move forward to the bicycle. Turn right, then move to the chair.
Turn left, and stop near the potted plant.



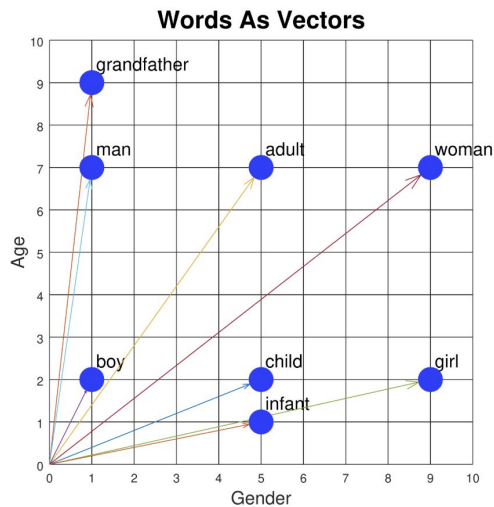
- What does *move forward* mean?
- What is a *bicycle*?
- How does it look?

— Robots need grounding

From NLP to Robotics

NLP:

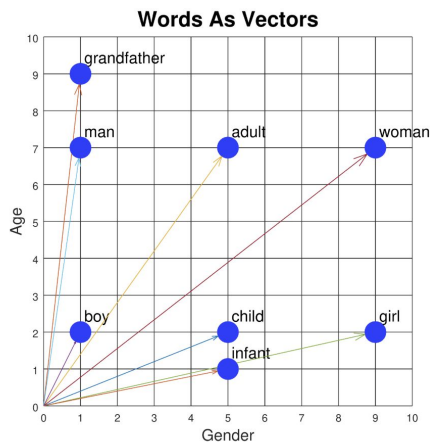
Text $\rightarrow$ Embeddings $\rightarrow$ Meaning



From NLP to Robotics

NLP:

Text $\rightarrow$ Embeddings $\rightarrow$ Meaning



Robotics:

Images $\rightarrow$ Perception $\rightarrow$ **Semantic Map** $\rightarrow$ Action
Language $\rightarrow$ Embedding $\rightarrow$

RGB

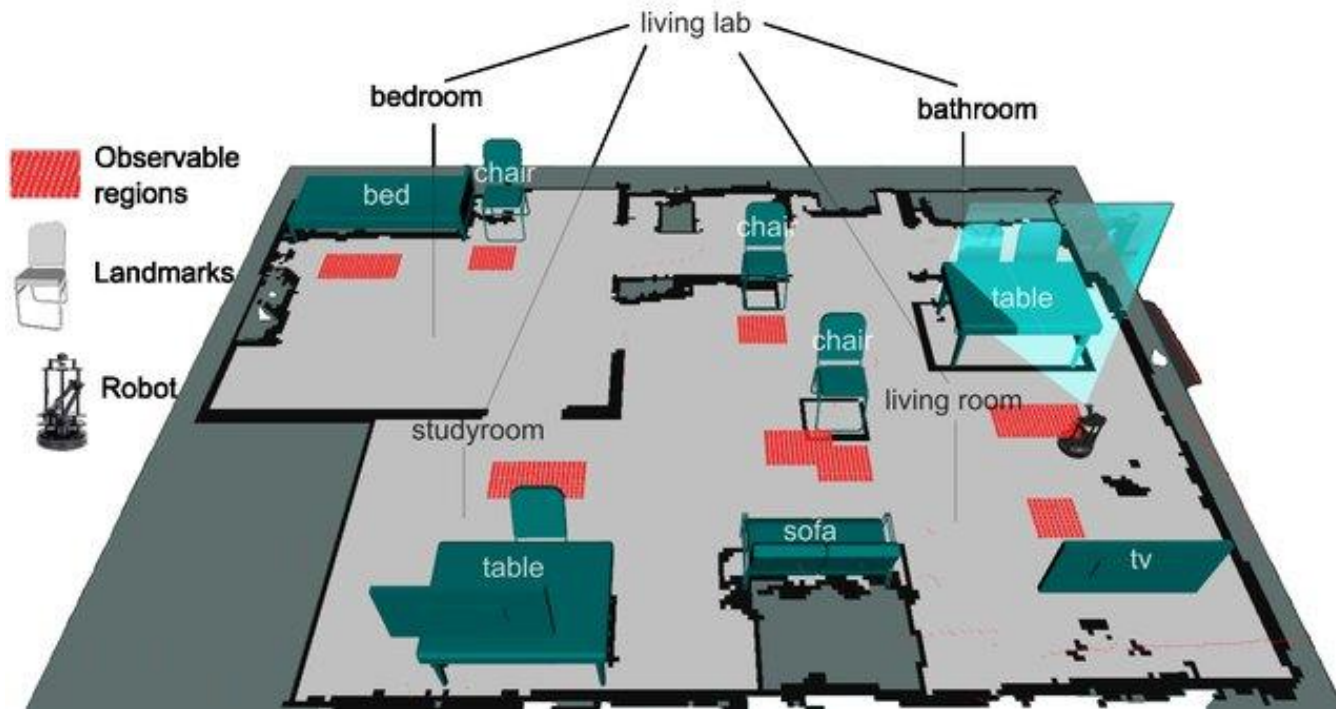


'Go to the chair'

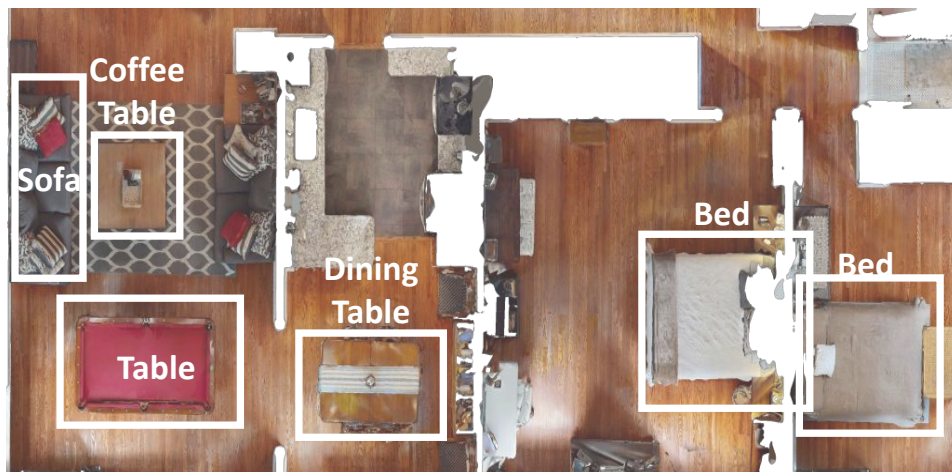


Action

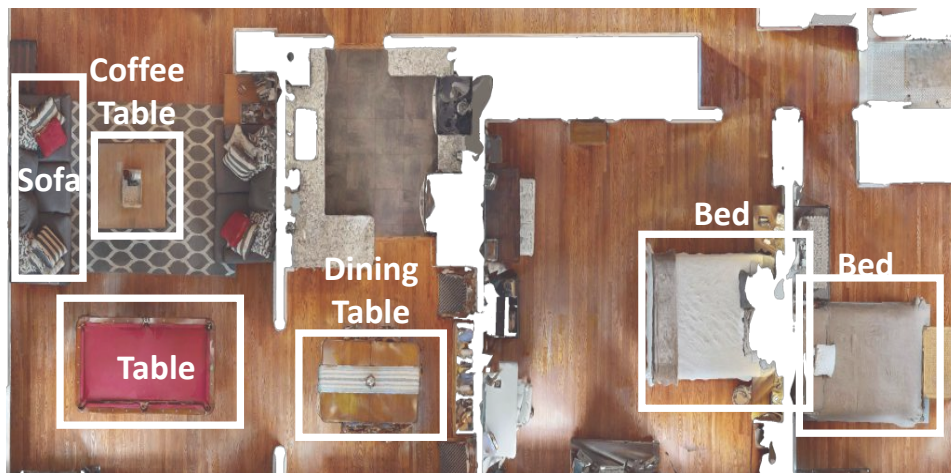
What is a Semantic Map?



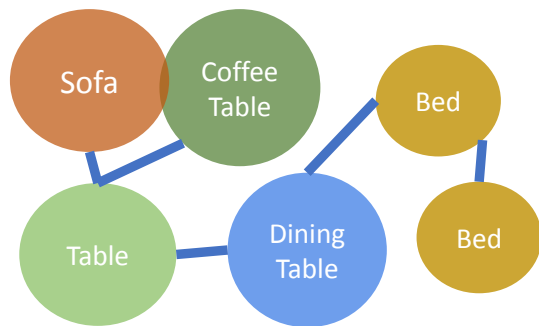
Map Structure



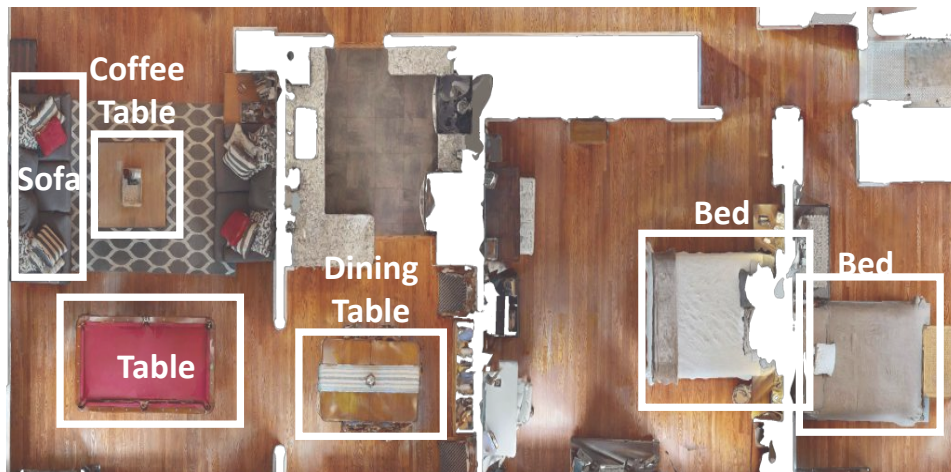
Map Structure



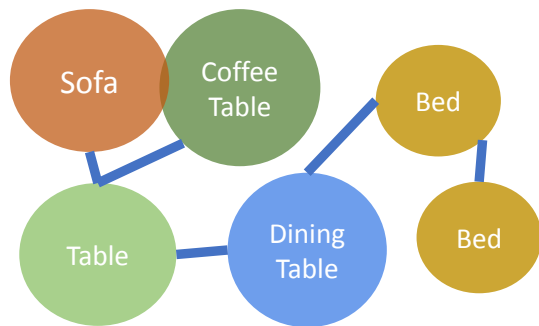
Topological Map



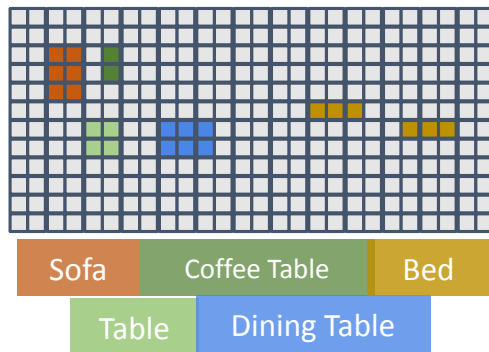
Map Structure



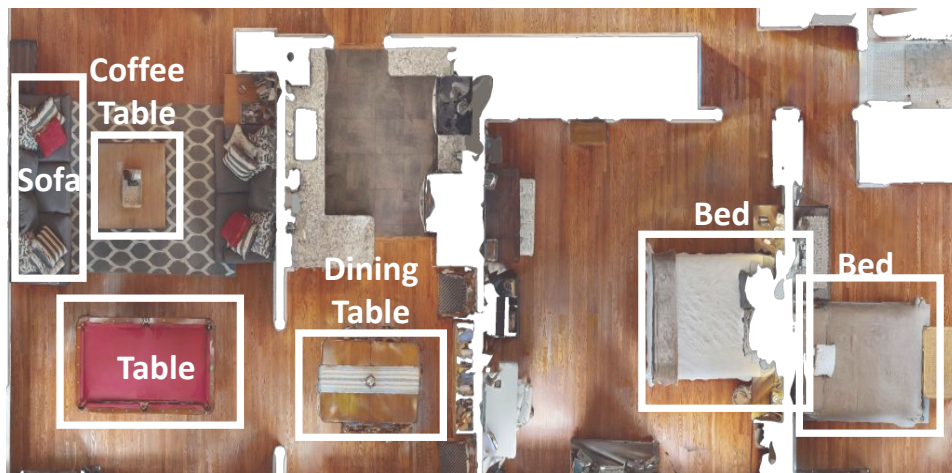
Topological Map



Grid Map



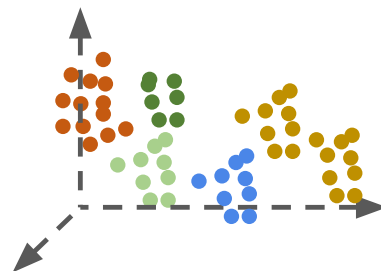
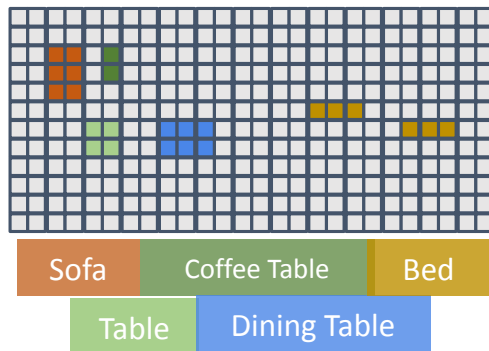
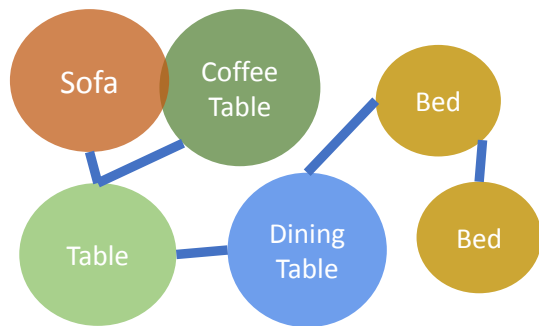
Map Structure



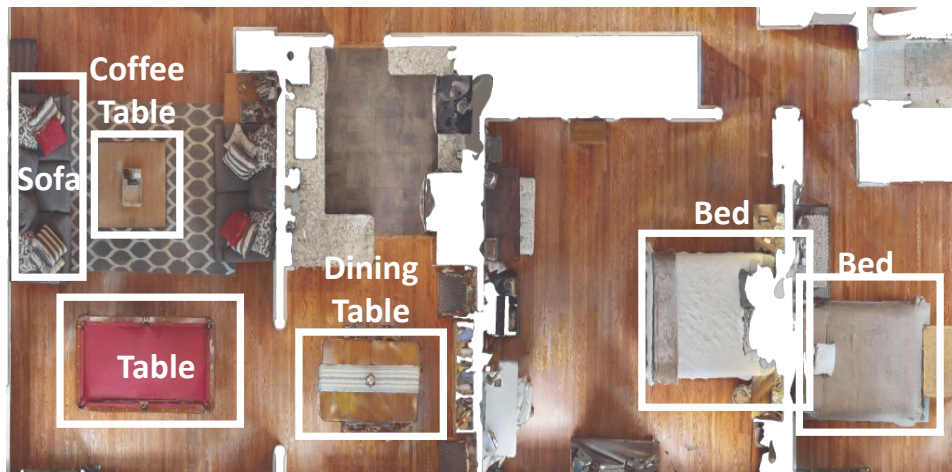
Topological Map

Grid Map

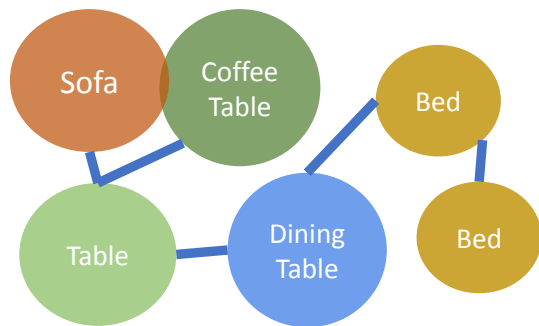
Geometric Map



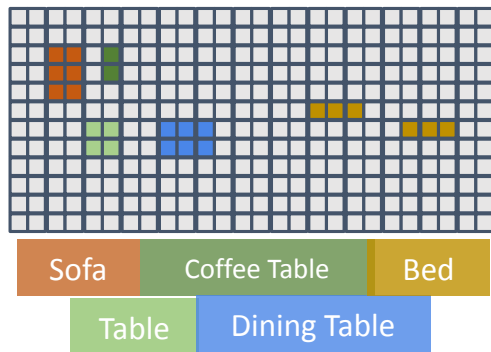
Map Structure



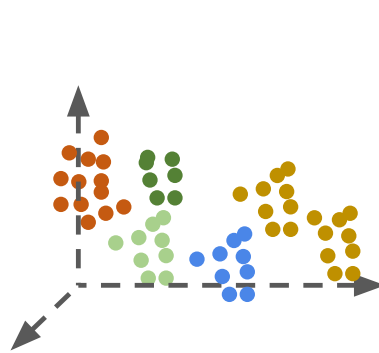
Topological Map



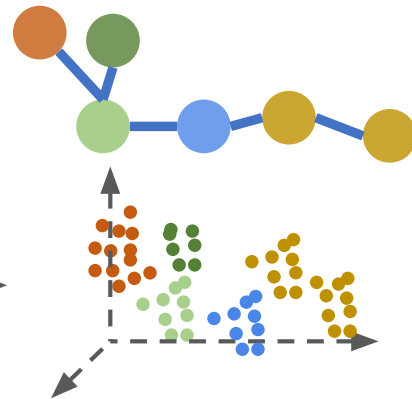
Grid Map



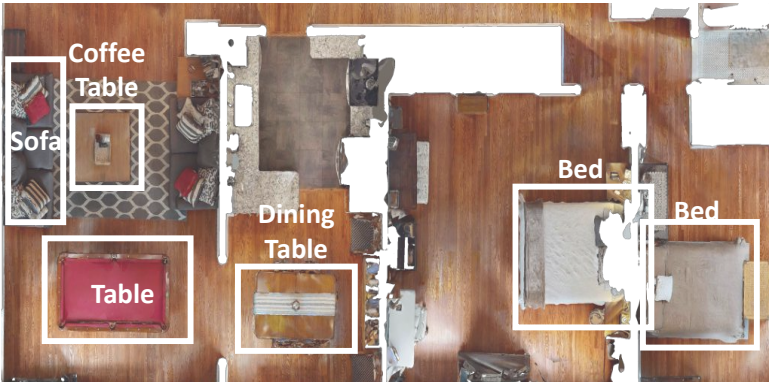
Geometric Map



Hybrid



Map Structure



Sparse

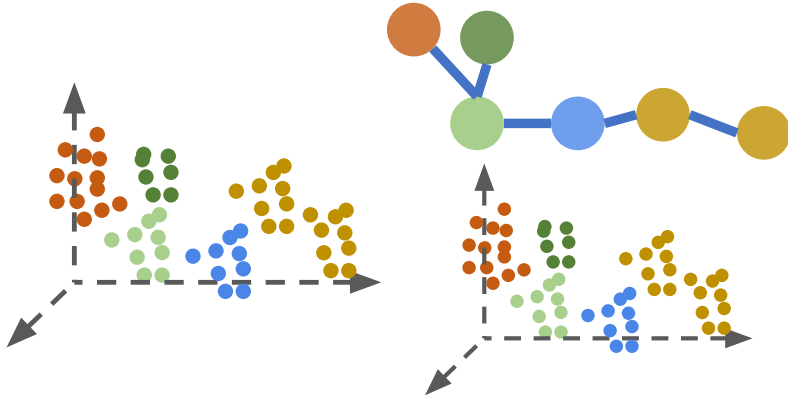
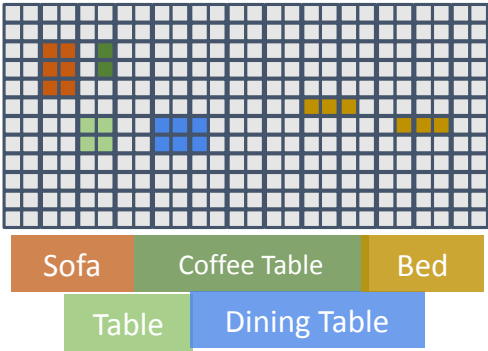
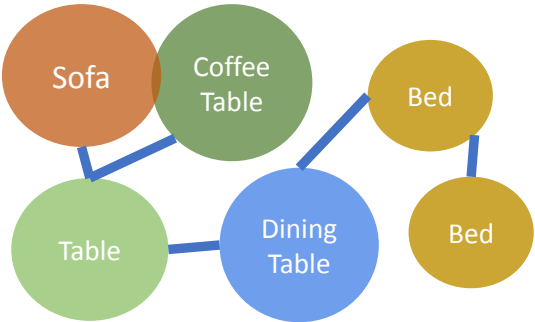
Dense

Topological Map

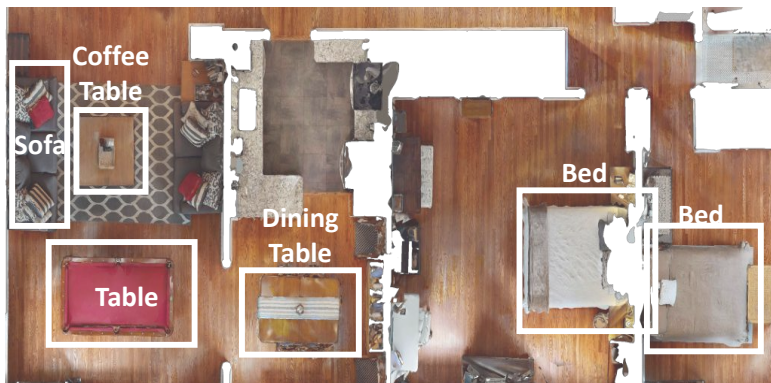
Grid Map

Geometric Map

Hybrid

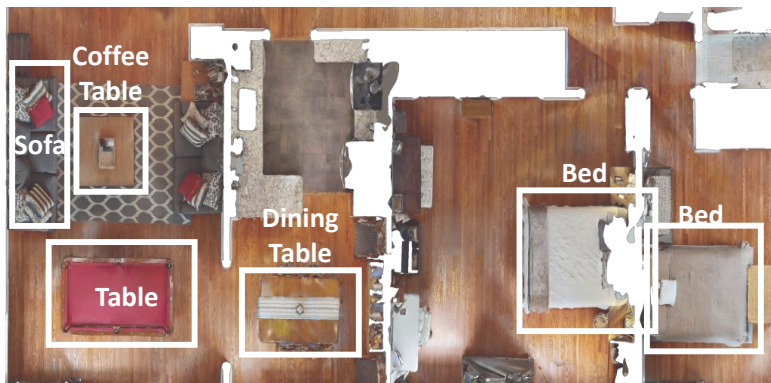


Map Encoding



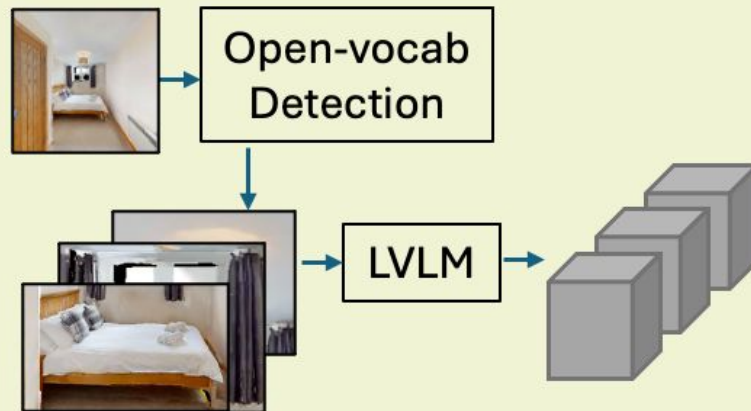
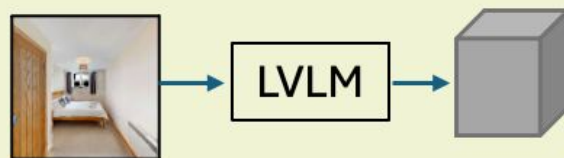
Explicit Encoding		
Occupancy	Binary	[0,1]
Image	RGB pixel values	
Objects	RGB pixel values	
	One-hot for n predefined object types	
	Text labels for object types	"bed"
	room types	
	Text labels for room types	"bedroom"
Audio Intensity	Single value in the range	[0-1]

Map Encoding



Implicit Encoding

Open Vocabulary



How It Enables Robot Tasks

goal becomes meaningful, not just geometric

Navigation
“go to kitchen”

Object search
“find mug”

Interaction
“pick up object”

Without semantics

- random exploration
- inefficient search

With semantics

- directed navigation
- intelligent search

Key Challenges

Why is this still an open research problem?

Language is Ambiguous

‘go to the orange scooter’



Dynamic Environments

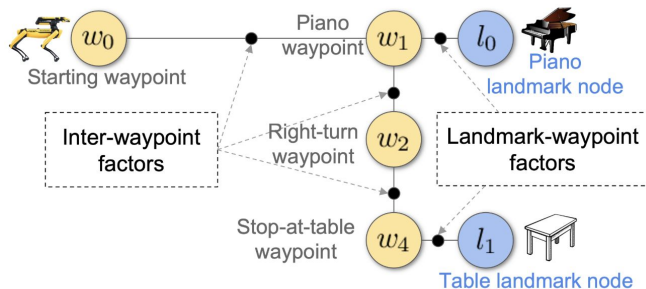
*mug on the table →
mug in sink*



Generalization



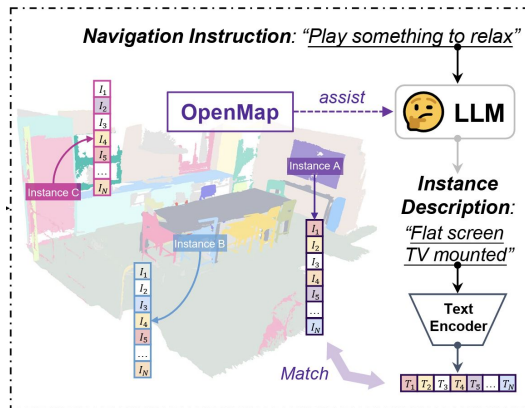
Language as Knowledge (LLMs for Robots)



Raychaudhuri et al. "Zero-shot Object-Centric Instruction Following: Integrating Foundation Models with Traditional Navigation." [2025]

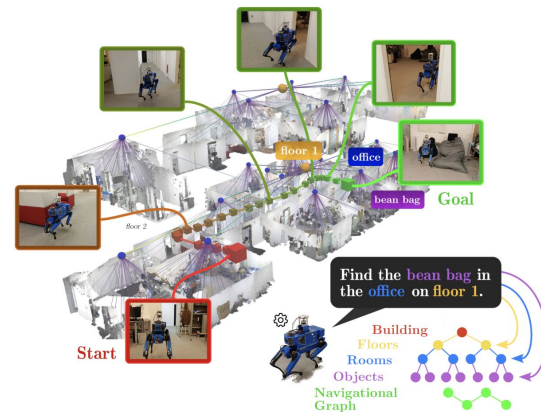
Future Directions

Open-World Understanding



Li et al. "OpenMap: Instruction Grounding via Open-Vocabulary Visual-Language Mapping." [2025]

Hierarchical Representations



Werby et al. "Hierarchical Open-Vocabulary 3D Scene Graphs for Language-Grounded Robot Navigation." [2024]

Takeaways

The next step for NLP is not just understanding text — but understanding the world
and semantic representations are the bridge between language and physical world

